



Power BI 360 powered by **twelve**¹²

Final Project

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Project Context and Objectives

This Power BI project was developed as the **final assignment** of the *Power BI 360* course taught by **Luis Mosquera Toscano** from *Twelve powered by Objetivo Analista*. The work aims to design and implement a **private club management space** for a grassroots football club, integrating **financial, sporting, and injury-related** information into a cohesive analytical environment. The project simulates a real-world data analytics scenario where data from multiple operational areas must be consolidated, modeled, and visualized to provide actionable insights to club management. The names of players are real and were extracted from **Transfermarkt**, while all other sporting, financial, and injury data were synthetically **generated using Claude AI**.

Project Goals

The main objectives of this project are:

- 1. **Centralize club data** related to finances, sporting performance, and player health into an integrated **Power BI data model**.
- 2. **Automate key performance metrics** through the creation of DAX measures and dynamic dashboards.
- 3. **Provide a comprehensive analytical tool** for decision-making across the club’s two teams, the *Youth Team* and the *Senior Team*.
- 4. **Develop an interactive reporting experience** with navigation menus, bookmarks, and dynamic visuals for improved usability.

Part Two | Data Modeling and Visualization

In Power BI, relationships were established between the datasets through the **Player_ID** and **Team_ID** keys, forming a **star-schema model** optimized for reporting and DAX calculations. A dedicated **Measures Table** was created to centralize all computed KPIs and metrics, including:

- **Financial Measures**
 - *Total Revenue Collected*
 - *Outstanding Payments*
 - *Payment Compliance Rate*
- **Sporting Measures**
 - *Total Goals, Assists, Cards, Matches Played*
 - *Goals per Match, Assists per Match*
 - *Goals per 90 Minutes, Assists per 90 Minutes, Cards per 90 Minutes*
 - *Minutes-Weighted Match Rating*
- **Injury Metrics**
 - *Total Injuries per Player*
 - *Average Recovery Time (Days)*
 - *Most Common Injury Type*

Dashboard Design and Pages

The Power BI report is composed of **three interactive pages**:

- 1. **Financial Dashboard**
 - Displays payment compliance, revenue breakdown by team, and pending amounts.
 - Allows filtering by team, payment status, and player.
- 2. **Sporting Dashboard**
 - Provides performance analysis per player and per team.
 - Includes metrics such as goals, assists, cards, minutes played, and per-90 rates.
 - Interactive slicers and player-level KPIs allow detailed performance exploration.
- 3. **Injury Dashboard**
 - Summarizes injury data using visualizations and a **Synoptic Panel** to identify the most frequent injury types.
 - Includes filters for injury type, player, and team.

Acknowledgment

Many of the analytical approaches, DAX logic structures, and visualization techniques implemented in this project were **inspired by and promoted by Luis Mosquera Toscano**, whose instruction and methodology throughout the *Power BI 360* course were fundamental to achieving a professional and analytical result. **Special thanks** are extended to him for his guidance, dedication, and the exceptional quality of the course content. Gratitude is also extended to **Twelve**, for fostering a stronger and more connected football analytics community, where **knowledge, collaboration, and data-driven learning** are central pillars for continuous growth and improvement within the sport.



FC Porto

Financial Information

Report

Context

Financial Information

Sporting Information

Injury Information

End

Team Name

Tudo

Payment Status

Tudo

Payment Method

Tudo

Total Revenue (Paid)

15 M

Payment Method

MBWay

€ 4,783,074

Cash

€ 4,586,751

Debit Card

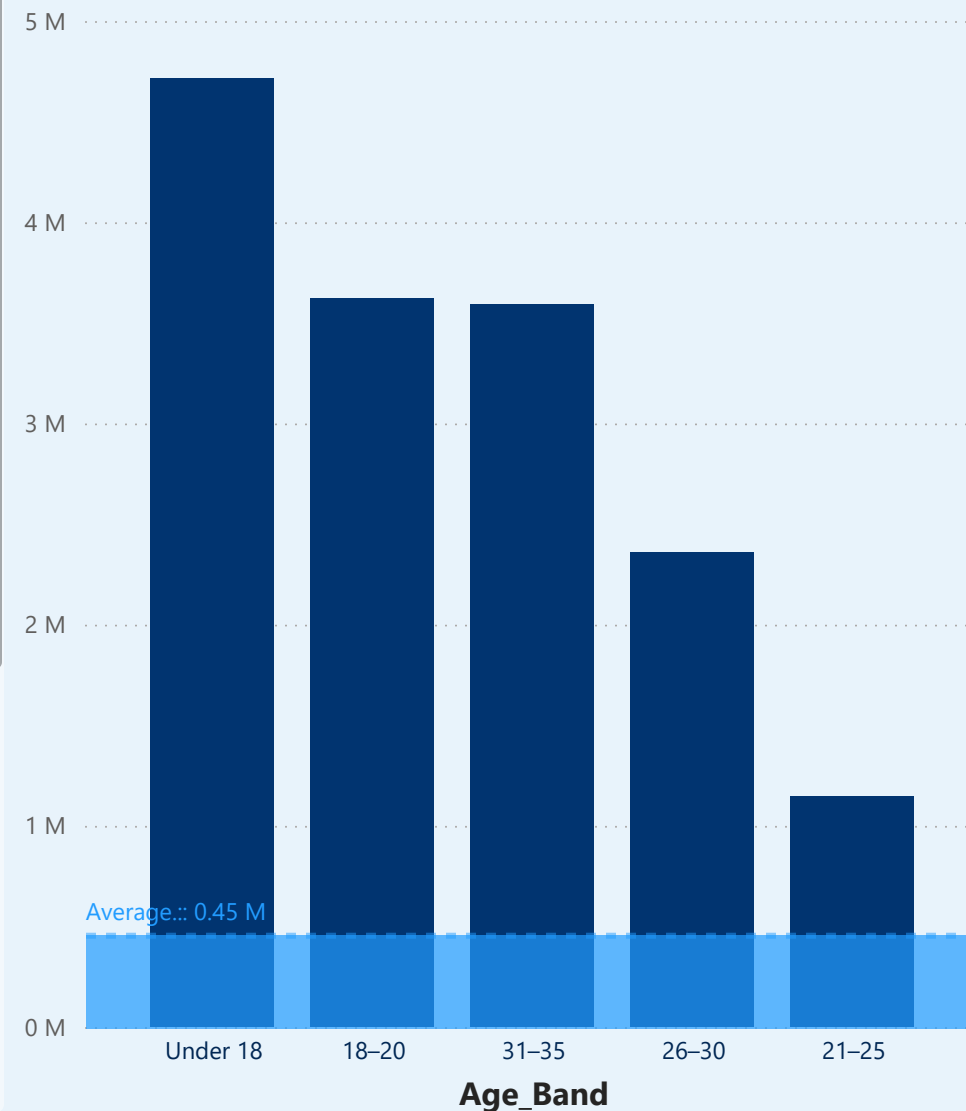
€ 3,773,520

Bank Transfer

€ 2,282,364

Team	Player	Annual Fee €	% of Total Revenue (Paid)
Senior Team	Alan Varela	€ 240,724	1.56%
Senior Team	André Miranda	€ 852,714	5.53%
Senior Team	António Ribeiro	€ 131,087	0.85%
Senior Team	Borja Sainz	€ 393,740	2.55%
Senior Team	Brayan Caicedo	€ 770,471	4.99%
Senior Team	Cláudio Ramos	€ 130,261	0.84%
Senior Team	Dinis	€ 721,703	4.68%
Senior Team	Filipe Sousa	€ 877,735	5.69%
Senior Team	Gabri Veiga	€ 974,531	6.32%
Senior Team	Gonçalo Ribeiro	€ 772,598	5.01%
Senior Team	Gonçalo Silva	€ 370,359	2.40%
Senior Team	Gonçalo Sousa	€ 393,815	2.55%
Senior Team	Jan Bednarek	€ 83,200	0.54%
Senior Team	João Costa	€ 513,182	3.33%
Senior Team	Kauê	€ 930,121	6.03%
Senior Team	Leonardo Fajardo	€ 438,111	2.84%
Senior Team	Luuk de Jong	€ 19,694	0.13%
Senior Team	Nehuén Pérez	€ 623,768	4.04%
Senior Team	Rui Monteiro	€ 177,712	1.15%
Senior Team	Samu Aghehowa	€ 921,937	5.98%
Senior Team	Tiago Andrade	€ 547,640	3.55%
Senior Team	Victor Froholdt	€ 128,692	0.83%
Senior Team	Zaidu	€ 985,542	6.39%
Youth Team	Alberto Costa	€ 580,908	3.77%
Youth Team	Ángel Alarcón	€ 674,647	4.37%
Youth Team	Anhá Cande	€ 48,677	0.32%
Youth Team	Bernardo Lima	€ 574,135	3.72%
Youth Team	Deniz Gül	€ 526,010	3.41%
Youth Team	Diogo Costa	€ 571,304	3.70%

Total Fee (Paid by Age) per Age Band





FC Porto

Sporting Information

Team Name

Youth Team



543

Total Matches Played

28 K

Total Minutes Played

118

Total Goals

0.37

Goals per 90

430

Total Starts

52.18

Minutes per Match

0.22

Goals per Match

0.18

Assists per Match

Pitch Analysis

Scatter Plot Analysis

Team	First_Name	Last_Name	Pos
Youth Team	Ángel	Alarcón	CB-Sub
Youth Team	Gabriel	Brás	CB-Sub
Youth Team	Kotaro	Nagata	CB-Sub
Youth Team	Bernardo	Lima	DM-Sub
Youth Team	Felipe	Silva	DM-Sub
Youth Team	Martim	Cunha	DM-Sub
Youth Team	Pedro	Lima	DM-Sub
Youth Team	William	Gomes	DM-Sub
Youth Team	Dominik	Prpić	GK-Sub
Youth Team	Trofim	Melnichenko	GK-Sub
Youth Team	Anhá	Cande	LW-Sub
Youth Team	Alberto	Costa	W-Sub
Youth Team	Gil	Martins	W-Sub

Report

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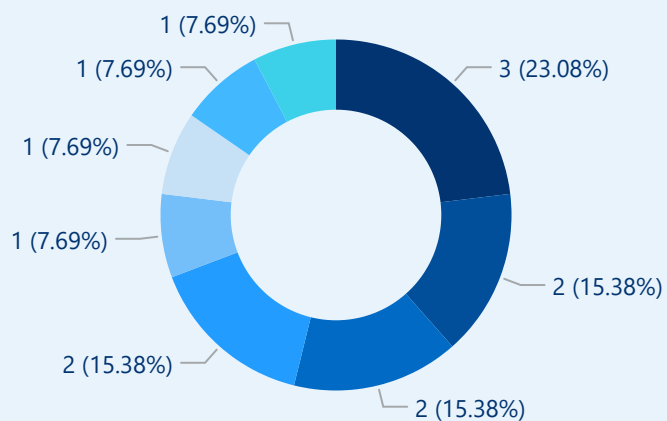


FC Porto

Injury Information

Team	Player	Body_Area	Days	Injury_Type	Severity Code
Youth Team	Alberto Costa	Brain	13	Cut/Laceration	●
Youth Team	Deniz Gül	Knee s	7	Concussion	●
Senior Team	Gabri Veiga	Thigh	19	Cut/Laceration	●
Youth Team	Gil Martins	Foot	21	Muscle Strain	▲
Senior Team	Kauê	Wrist	173	Fracture	◆
Youth Team	Kotaro Nagata	Brain	51	Fracture	▲
Youth Team	Kotaro Nagata	Shoulder s	20	Muscle Strain	●
Senior Team	Leonardo Fajardo	Thigh	27	Bruise	▲
Youth Team	Pablo Rosario	Brain	19	Fracture	●
Senior Team	Rui Monteiro	Knee s	108	Fracture	◆
Senior Team	Rui Monteiro	Leg Upper	175	Fracture	◆
Senior Team	Samu Aghehowa	Leg Upper	165	Cut/Laceration	◆
Youth Team	William Gomes	Head Soft Tissue	14	Ligament Sprain	●

Body Area Injuries



Body_Area

- Brain
- Knee s
- Leg Upper
- Thigh
- Foot
- Head Soft Tissue
- Shoulder s
- Wrist

Average Injury Days

62.46



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